

SAFER 247: A RADIO FREQUENCY-BASED EMERGENCY MESSAGING SYSTEM

CHAPTER I

INTRODUCTION

1. Statement of the Problem

The MDRRMO of San Fernando, Bukidnon maintains effective emergency response operations, however its current radio-based communication setup is limited to voice transmission and does not support the structured transmission of digital emergency data over radio frequency. This allows basic verbal communication but limits sending of structured emergency information such as location coordinates, situation reports, and images needed for accurate and timely decision-making during incidents. In addition to voice transmission, the MDRRMO does not have a dedicated system to send and receive digital emergency data through radio frequency and a centralized command center application to visualize and organize incoming field reports. Therefore, this study aims to develop a Radio Frequency-Based Emergency Messaging System for the DRRMO of San Fernando, Bukidnon.

Specifically, this study aims to answer the following problems:

1. How can a mobile and web-based radio frequency emergency messaging module be designed and developed to enable the encoding and decoding of text, geolocation, and image data through audio modulation?
2. How can a web-based command center dashboard be designed and implemented to enable the geospatial visualization of received emergency reports through incident mapping and heatmap representation?
3. How can an image processing module be developed to enhance the quality and clarity of images received after radio frequency transmission?
4. How can a responder tracking module be developed to enable field responders to beacon their geographic location to the command center?

5. How can the developed system be evaluated in terms of functionality, transmission and reception performance, image enhancement quality, and perceived usability among intended end-users?

2. Objectives of the Study

The study aims to design and develop an emergency communication system that enables transmission and reception of emergency information through radio frequency communication.

Specific Objectives:

1. To design and develop a mobile and web-based radio frequency emergency messaging module that enables the encoding and decoding of text, geolocation, and image data using audio modulation.
2. To design and implement a web-based command center dashboard that enables the geospatial visualization and organization of received emergency reports through incident mapping and heatmap representation.
3. To develop an image processing module that enhances the quality and clarity of transmitted images.
4. To develop a responder tracking module that enables responders to beacon their geographic location to the command center.
5. To evaluate the developed system based on functionality, transmission and reception performance, image enhancement quality, and perceived usability among intended end-users.

3. Scope and Limitation of the Study

This study will cover the development of an emergency communication system using radio frequency, which will be used when cellular networks or internet connectivity fail. It will support responder-to-responder and responder-to-command-center emergency communication for Disaster Risk Reduction and Management Offices (DRRMOs). The system will involve an Android

mobile application for responders and a web application for command center use. The mobile application will provide a messaging module and a location tracking module, while the web application will feature a messaging module, dashboard module, location tracking module, and an image enhancement module for received radio-transmitted images.

The messaging module will support the transmission and reception of text, geolocation, and image data through radio devices over radio frequency bands via acoustic coupling and direct audio cable interfacing. Text and geolocation data will be encoded into digital audio signals using APRS, while image data will be converted into analog audio tones using SSTV and then decoded back into their respective digital formats on the receiving side. The module will support two types of messages: incident reports, which will contain the type of incident being reported and its corresponding incident description, and quick messages which may contain text, geolocation, or an image. Sender identification and geolocation data will be automatically embedded in every message regardless of type. Additionally, to ensure no external parties may gain access to confidential information sent over radio, senders will have the option to hash their messages before being encoded into audio tones; to access the original message, the receiving end must have access to a shared secret key.

It should be noted that the reliability and quality of received messages will depend on the radio hardware being used, and the signal coverage will be limited to the operational range of available repeater stations. The accuracy of the geolocation coordinates will also be dependent on the GPS capability of the device used and may differ depending on signal strength and environmental conditions such as being indoors or in areas where satellite visibility is poor. The module will not support voice or video communication, and file transfers beyond image transmission. It will not implement data encryption or security protocols, and will remain limited to the broadcast nature of radio frequency communications. Furthermore, because analog radio only allows one transmission at a time, responders cannot simultaneously send incident reports. The command center has to decide how to organize and prioritize incoming reports.

The dashboard module will support geospatial visualization and organization of received incident reports. It will contain a map view of incident reports, and a heatmap view to show areas where incidents are frequently occurring. Additionally, the module will provide distribution charts to visualize the number of incidents per type and per barangay, and an incident timeline chart to visualize when incidents frequently occur. Lastly, it will list all received incident reports in a reports panel. Resource allocation, responder dispatching, and responder monitoring will be outside the scope of this module.

The location tracking module will utilize the APRS protocol to periodically transmit responder locations to the command center. The responder mobile application will have a beaconing feature that can be enabled or disabled as needed. When enabled, the application will transmit the current coordinates of the responder to the command center in five-minute intervals. The command center web application will have a tracking interface which will display last-known responder locations. The beaconing of location will have to be manually enabled or disabled because the mobile application has to continuously listen for incoming messages in its messaging module, turning on the beacon would mean it cannot receive messages or reports.

The image enhancement module will perform a pipeline of classical image processing techniques to improve the quality of received SSTV images. It will not run automatically upon reception and will only perform when explicitly requested by the user. This module will only be available on the command center since it will require significant computing power and battery usage, both of which would not be ideal for the responder side.

CHAPTER III

TECHNICAL BACKGROUND

1. System Architecture of the Proposed System

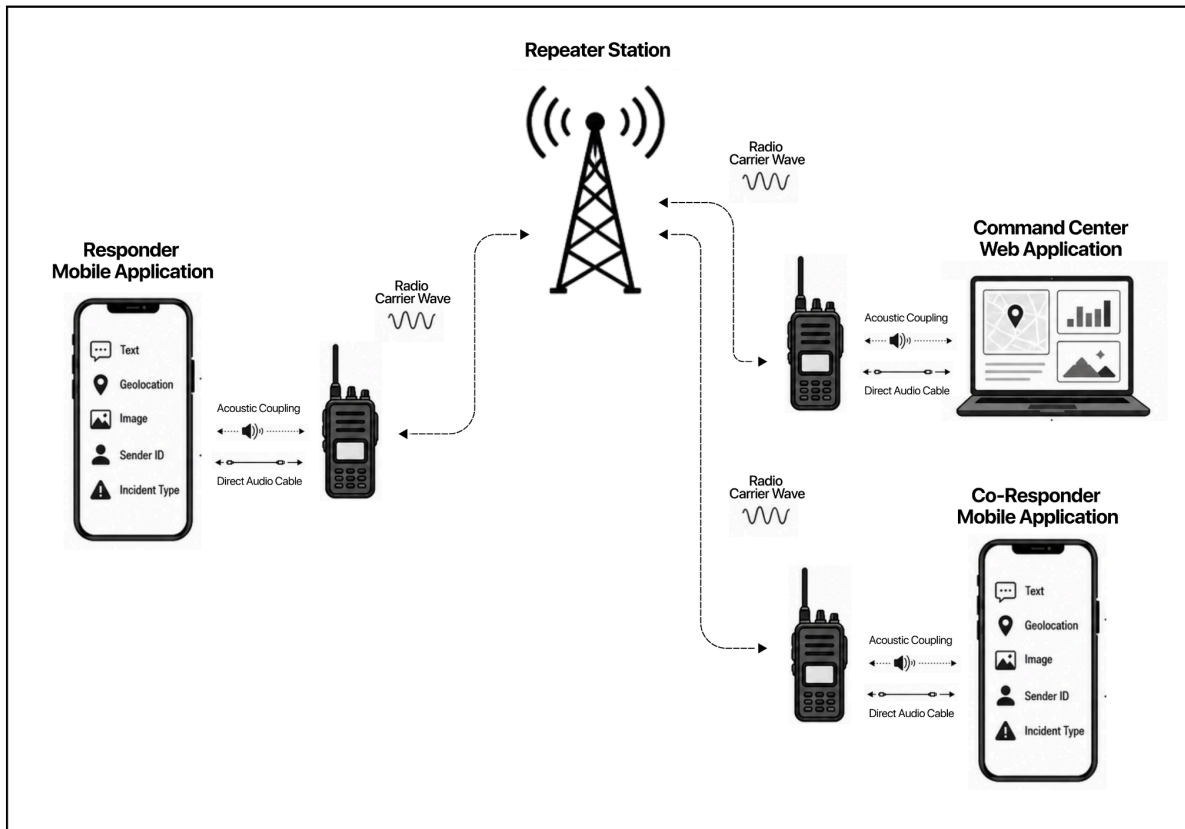


Figure 1. System Architecture of the Proposed RF-Based Emergency Messaging System

As for the architecture of this system, it is based on RF communication that would enable the transmission of emergency information between field responders and the command center. Figure 1 illustrates this, showing how data originated in the responder mobile application is encoded, transmitted through RF, and received by the receiving units. The communication between the field units and the command center would be carried out within the DRRMO of San Fernando, Bukidnon.

On the field-side communication, it starts in the responder mobile application. In this application, a responder would be able to submit emergency information in the form of text, messages, location, images, sender identification, and incident type. However, before transmission, the data must be first encoded into audio signals since the communication takes place over RF. The transfer of audio signals from mobile device to radio transceiver may be implemented through acoustic coupling, which uses the speaker and microphone of devices, or direct audio cable interface.

Once the audio signal is sent to the transceiver, it will then be transmitted over RF. The repeater station picks up the signal and retransmits it to other units on the same channel. In the current setup, co-responders serve as parallel receivers, meaning that they receive and decode the same message submitted to the command center at the same time.

On the receiver side, both co-responders and the command center need radio transceivers to receive the transmitted signal. For the co-responder, after passing through the radio transceiver, the audio goes into the mobile application where it is decoded and presented in digital format. For the command center, the decoded data will be used not only for emergency response but also for the web application for the creation of incident visualizations.

The described architecture enables the transmission of digital emergency information over RF without the use of cellular or internet connection. The mobile application encodes the data, transceivers and repeaters transfer the data, and the receiving application decodes and presents it.

2. Supporting Technologies and Tools

2.1. Java

Java is a high-level, object-oriented programming language that is executed using the Java Virtual Machine (JVM) platform. This allows the program to run on multiple platforms, since JVM is compatible with different operating systems. Java is the primary programming language used for Android development by means of the Android Software Development Kit (*Java Documentation - Get Started*, 2023). Java

is chosen as the programming language for this project, since it will be used to develop the Android mobile application for field responders. Java will be responsible for handling user interface, audio input/output for radio communication, GPS data collection for automated geotagging, and message composition and transmission process.

2.2. Python

Python is a general-purpose programming language commonly used for its easy-to-read syntax and extensive library support, ranging from web development, scientific computing, and machine learning (The Python Language Reference, n.d.). In this project, Python will be used as the primary language used in the command center web application as well as the image processing enhancement algorithm. In the case of the web application, Python will control the server-side of the program, responsible for receiving and decrypting any transmitted messages and presenting the monitoring interface to the user. Python will be used to implement the classical image enhancement pipeline using OpenCV, Pillow, and related image processing libraries.

2.3. C++

C++ is a programming language that provides a mix of direct memory manipulation capabilities and higher-level abstractions, which allows it to be used for high-performance applications (Stroustrup, 2013). In this project, C++ will be used for all critical signal manipulations and transformations such as encoding and decoding using APRS and SSTV algorithms. Such tasks are resource-intensive due to numerous calculations required, such as FFT computations and audio tone generation as well as the sequential decoding process. In order to implement these processes effectively, C++ will be used because of its fast execution and ability to access hardware directly. The resulting C++ library will be linked with both the Android application and the Python back-end application via NDK and Python native extensions.

2.4 Automatic Packet Reporting System (APRS)

Automatic Packet Reporting System (APRS) is a real-time digital communications protocol developed for amateur radio to transmit tactical data such as GPS coordinates, telemetry, text messages, and alerts over radio frequencies (Christanto et al., 2025). APRS uses the AX.25 packet radio protocol to structure digital information into standardized packets, which are then modulated into audio signals using Audio Frequency Shift Keying (AFSK) for radio frequency transmission. In this study, it will be used for the encoding and decoding of text, location, and incident reports in the emergency messaging module, and for the location tracking module.

2.5 Slow Scan Television (SSTV)

Slow Scan Television (SSTV) is the technique of transmitting still images using radio waves through encoding an image into a series of audio signals, which are transmitted using modulation of a carrier wave. Once the audio signals are captured on the receiving end, it will then be decoded to rebuild the original image (Ok, n.d.). The use of SSTV in this case is justified by its capability to operate within the limited bandwidth of regular amateur radio communication channels, hence enabling the transmission of images using ordinary radio communication equipment without requiring any special equipment. There are various SSTV modes, such as Robot 36, Scottie, and Martin, and each of these modes offers different advantages concerning transmission speed, resolution, and color depth. In this study, SSTV will be used for image transmission because of its audio modulation, which allows for the effective transmission of signals using the acoustic and direct coupling techniques.

2.6 SQLite

SQLite is a simple, serverless database management system that stores all data in one file only (SQLite Home Page, n.d.). Unlike other client-server databases, it does not require any server process to be installed separately, and this is precisely what makes SQLite suitable for mobile applications. It offers the ability to use

standard SQL, and guarantees persistence of data without having to deal with the extra intricacies of network-based systems. The SQLite application will be used in this project to store the message history and transmission information on the responder's device, making it possible to review past transmission information on their mobile device.

2.7 PostgreSQL

PostgreSQL is an open source Relational Database Management System that stores and manages data (PostgreSQL, 2026). It will be the main database for the command center web application and will contain emergency message information, sender information, what type of incident, and geographic information on the location. This tool may be used in conjunction with the PostGIS extension; this will prove to be very useful for dealing with geospatial data. The monitoring dashboard will be crucial since all incidents need to be plotted on a map, and PostGIS will enable the system to store and index geographic coordinates.

2.8 HMAC-SHA256

Hash-based Message Authentication Code with SHA-256, or HMAC-SHA256, is a cryptographic algorithm that verifies both the integrity and authenticity of a message by combining a cryptographic hash function with a shared secret key to produce a fixed-length authentication tag unique to both the message content and the key (Rahim et al., 2022). Any alteration to the message during transmission causes the tag to fail verification at the receiving end, making it effective for detecting tampering or corruption. In the proposed system, HMAC-SHA256 will be used to authenticate APRS-encoded transmissions at the packet level, where each packet will include an HMAC tag computed from the payload and a pre-shared key known to both devices. When a packet is received, the system will recompute the tag using the same key and compare it against the received one, rejecting any packet where the tags do not match.

CHAPTER IV

METHODOLOGY

1. Conceptual Framework

For the proposed system, there will be two major components, namely the responder mobile application and the command center web application. The responder mobile application will act as the main communication application in the field, allowing emergency responders to record and send data about the incident, including geolocation data, text emergency reports, and images. This application will have an encoding and decoding process for converting the recorded data into audio data that can be sent through RF. Additionally, there will be a database for storing the received and transmitted data and images. It will also include a location tracking module where responders can beacon their location to the command center at five-minute intervals.

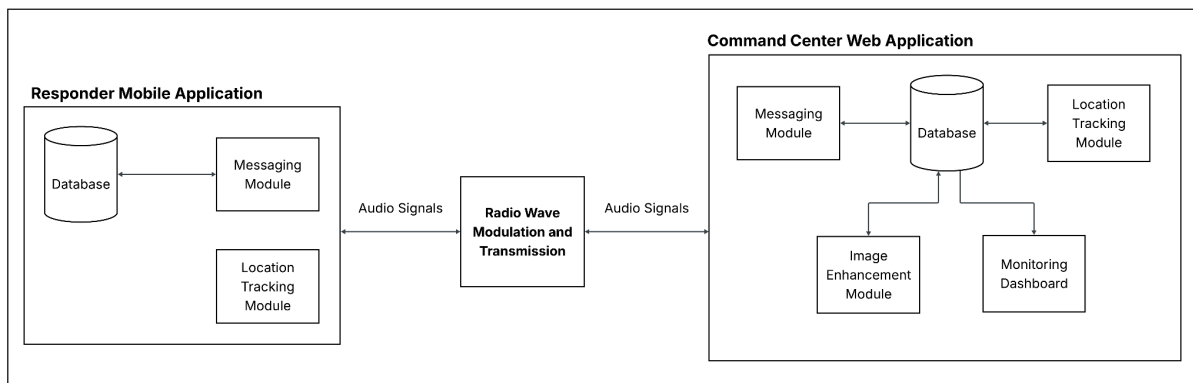


Figure 2. Conceptual Framework of the Proposed RF-Based Emergency Messaging System

The command center web application will act as the centralized coordination platform for emergency response activities. It will consist of the emergency response messaging module, image enhancement module, monitoring dashboard, and location tracking module. The messaging module will be responsible for executing

similar communication tasks as those of the responder application. Incident data collected will be saved into the database and displayed through the monitoring dashboard through an incident map and heatmap, as well as frequency and distribution charts. The responder tracking module will display the latest transmitted location of field responders. Finally, the image enhancement module will be responsible for enhancing the quality of the received images due to low-bandwidth radio transmission limitations.

2.1 Requirements Gathering and Analysis

2.1.1 Functional Requirements

A. Field Responder

Field responders are the emergency personnel deployed in the field who will use the Android mobile application for responder-to-responder and responder-to-command-center communication. The system shall allow field responders to:

1. Access the shared broadcast messaging channel for sending, receiving, and viewing emergency messages.
2. View a dedicated incident reports tab within the broadcast messaging channel.
3. Create and send quick messages containing text, geolocation, or image data.
4. Create and send incident reports containing incident type, incident description, and optional image attachment.
5. Receive quick messages and incident reports from other field responders and the command center.
6. View sent and received messages in the broadcast messaging channel.
7. Search messages in the broadcast messaging channel.

8. Filter messages according to incident type.
9. Filter messages according to date.
10. Edit their call sign, which shall serve as sender identification in outgoing messages.
11. Automatically attach sender identification, geolocation data, and timestamp in every outgoing message.
12. Resend failed messages without recomposing the same message details.
13. Send emergency information using a radio device connected by acoustic coupling or direct audio cable.
14. Enable or disable responder location beaconing to periodically transmit location to the command center.
15. Use the system when cellular network or internet connection is unavailable.
16. Continue receiving and decoding incoming RF transmissions while the application is running in the background, allowing the responder's device to remain active as a listener even when the screen is off or other applications are in use.

B. Command Center Operator

The command center operator refers to the DRRMO personnel assigned to monitor and manage received emergency information through the web-based command center application. The system shall allow command center operator to:

B.1 Broadcast Messaging Channel

1. Access the shared broadcast messaging channel.
2. Send quick messages to field responders.

3. Receive quick messages and incident reports transmitted by field responders.
4. View sent and received messages in the broadcast messaging channel.
5. View a dedicated incident reports tab within the broadcast messaging channel.
6. Search messages in the broadcast messaging channel.
7. Filter messages according to incident type or date.
8. Resend failed outgoing messages without recomposing the same message details.
9. Automatically attach sender identification, geolocation data, and timestamp in every outgoing message.
10. View incident reports in a dedicated incident report sub-channel, which is logically separated from the general broadcast messaging feed.

B.2 Dashboard

1. View recent incident reports.
2. View distribution charts showing the number of incidents per incident type.
3. View distribution charts showing the number of incidents per barangay.
4. View monthly incident frequency charts.
5. View daily incident frequency charts.
6. Filter dashboard charts according to incident type.
7. Filter dashboard charts according to date when applicable.

B.3 Map View

1. View incident report markers according to their recorded coordinates.

2. Select an incident marker to view related incident information.
3. View clustered incident reports according to geographic proximity.
4. View a heatmap layer showing areas where incidents frequently occur.
5. View the latest beaconed location of the field responders.

B.4 Report Records

1. View the list of all received incident report records.
2. Filter incident report records according to date, incident type, or location.
3. View the details of a selected incident report.
4. View a selected incident report on the map.
5. Export incident reports with filters by date, incident type, and location which can be downloaded as .xlsx, .csv, or .pdf files.

B.5 Image Enhancement

1. View images received through RF transmission.
2. Select a received image from the broadcast messaging channel.
3. Manually apply image enhancement to a selected received image.
4. View the original and enhanced version of the image.

C. RF Communication Module

The RF communication module will handle the encoding and decoding of emergency messages through digital audio modulation, to be then transmitted through radio carrier waves. The system shall:

1. Encode text and geolocation data into digital audio signals using APRS.
2. Convert image data into analog audio tones using SSTV.

3. Interface with external radio devices through acoustic coupling or direct audio cable connection for transmitting and receiving encoded audio signals.
4. Decode received audio signals back into readable text, geolocation, or image data.
5. Support responder-to-responder and responder-to-command-center communication.
6. Support periodic beacon transmission for responder location tracking.
7. Store previously encoded messages and allow users to retransmit them without manually typing the contents again via a resend button.
8. Continue receiving and decoding incoming RF transmissions while application is running in the background.
9. Operate without relying on cellular data or internet connectivity during RF transmission.

D. Image Enhancement Module

The image enhancement module will improve the quality of received RF-transmitted images on the command center side. The system shall:

1. Accept a received RF-transmitted image as input.
2. Allow the command center operator to manually apply image enhancement.
3. Apply a pipeline of classical image processing techniques to improve the image quality.
4. Generate an enhanced version of the received image.
5. Preserve the original received image for comparison and reference
6. Allow the command center operator to view both the original and enhanced versions of the image

7. Store the enhanced image output.
8. Perform image enhancement only on the command center side.

2.1.3 Output Requirements

A. Field Responder

The field responder's emergency communication mobile application shall produce the following outputs:

1. Broadcast Message Feed - A display of sent and received emergency messages in the shared broadcast channel.
2. Sender Identification Display - The call sign of the sender attached to each message.
3. Geolocation Display - The location data that comes with each message.
4. Timestamp Display - The date and time of each sent or received message.
5. Resend Status Indicator - A message indicating whether the previously failed message was successfully resent or not.
6. Search Results - A filtered display of messages that matches the search input.
7. Filtered Message List - A display of messages filtered according to incident type or date.
8. Decoded Received Data - The comprehensible text, geolocation, or image data decoded from RF audio signals.

B. Command Center

The command center web application shall produce the following outputs:

1. Broadcast Message Feed - A display of sent and received emergency messages in the shared broadcast channel.

2. Sender Identification Display - The call sign of the sender attached to each message.
3. Geolocation Display - The location data embedded in each message.
4. Timestamp Display - The date and time associated with each sent or received message.
5. Search Results - A display of messages or transmissions matching the search input.
6. Filtered Broadcast Messages - A display of messages filtered according to incident type or date.
7. Recent Incident Reports Display - A dashboard section showing the latest received incident reports.
8. Incident Type Distribution Chart - A chart showing the number of incidents per incident type.
9. Barangay Distribution Chart - A chart showing the number of incidents per barangay
10. Monthly Incident Frequency Chart - A chart showing incident frequency per month.
11. Daily Incident Frequency Chart - A chart showing incident frequency per day.
12. Map Markers - Geospatial markers representing incident reports based on embedded location coordinates.
13. Clustered Incident Display - A map view of groups of incident reports according to geographic proximity.
14. Heatmap Visualization - A visual layer showing areas with frequent and seldom incident reports.
15. Responder Tracking Display - A map displaying the latest locations of field responders.

- 16. Report Records List - A list of all received incident reports.
- 17. Filtered Report Records - A report list filtered by date, incident type, or location.
- 18. Incident Report Details - A detailed view of a selected incident report.
- 19. Incident Report Export Data - The selected incident data for export which can be downloaded as .xlsx, .csv, or .pdf files.
- 20. Original Received Image Display - The Image received in RF without being processed.
- 21. Enhanced Image Display - The enhanced image produced by the image enhancement module.

2.1.5 Software Requirements

A. Field Responder

The responder mobile application shall require the following software components:

- 1. Android operating system
- 2. Android mobile application
- 3. SQLite for local message storage
- 4. Required Android permissions for location, camera, storage, or media access.

B. Command Center

The command center web application shall require the following software components:

- 1. Compatible web browser
- 2. Python-based web application environment
- 3. PostgreSQL database

4. PostGIS extension for geospatial data support
5. Image processing libraries for classical image enhancement
6. Software support tools for receiving and processing audio input from connected radio equipment

C. Development Tools

The researchers shall use the following development tools:

1. Android Studio for developing the responder mobile application
2. Visual Studio Code for developing the command center web application and supporting modules
3. Java for Android application development
4. Python for command center backend processing and image enhancement
5. C++ for RF encoding and decoding activities
6. SQLite for local storage in Android application
7. PostgreSQL with PostGIS for command center storage and geospatial data handling
8. OpenCV, Pillow, and other image processing libraries for image enhancement

2.1.6 Hardware Requirements

A. Field Responder

1. Android smartphone for running the emergency communication mobile application
2. Built-in GPS sensor for geolocation acquisition
3. Built-in microphone and speaker for acoustic coupling
4. Radio transceiver for RF transmission and reception

5. Audio cable or interface connector for direct audio cable interfacing

B. Command Center

1. Laptop or desktop computer for running the command center web application
2. Radio transceiver for RF transmission and reception
3. Microphone or audio input device for receiving audio from the radio device
4. Speaker or audio output device for sending encoded audio to the radio device
5. Audio cable or interface connector for direct audio cable interfacing

2.3 Prototype Development

2.3.2 Command Center Dashboard

The monitoring dashboard of the command center web application will display the incident reports coming from the messaging module. The geographical visualization of the dashboard will display the incident reports through an incident map along with heatmap visualization. Additionally, statistical visualization will display the incident reports by frequency and distribution. Overall, this dashboard aims to provide a system that goes beyond communication and will support situational awareness during emergency response.

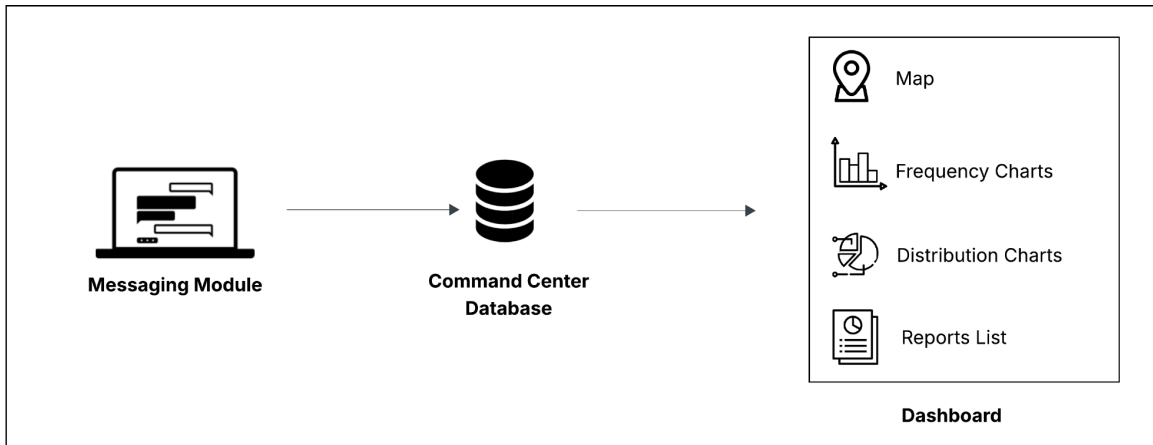


Figure 26. Command Center Dashboard Data Flow

2.3.2.1. Data Preprocessing

The incident reports received by the messaging module are stored in the system database. To use this data for visualization, the dashboard module will query the database for the data and will then perform data preprocessing methods to standardize and prepare the data for visualization. The relevant data that the dashboard module requires are the following: message ID, responder call sign, incident type, message content, latitude, longitude, and timestamp. This data will be structured in a format consistent with the requirements of the data visualization library Leaflet.js which is in the form of a GeoJSON object.

2.3.2.2. Geospatial Visualization

The structured data will then be consumed by the geospatial visualization components in the dashboard. For the incident map, the latitude and longitude of each report will be extracted to create a map marker on an interactive map. Each marker represents an incident and when clicked displays the full incident details including sender call sign, incident type, message content, and timestamp.

For heatmap visualization, the dashboard will retrieve the latitude and longitude coordinates of all received incident reports. Each incident report will contribute an equal weight to the visualization, allowing the spatial density of reports

to be based on the concentration of nearby geographic points. Locations with higher concentrations of reported incidents will be displayed with greater color intensity, enabling command center personnel to rapidly identify areas experiencing a large number of emergency reports and prioritize situational assessment and response. Additionally, when a region or barangay is clicked, it will display the related reports in that region which can then be filtered and sorted

2.3.2.3. Statistical Visualization

For the statistical visualization of emergency incidents, the dashboard module will retrieve the incident report dataset from the database and aggregate the reports according to the requirements of each statistical visualization. The timestamp associated with each report will be used to generate the incident timeline and to group incidents by hour, day, and month for the hourly, daily, and monthly incident frequency charts. Similarly, the incident type field will be used to determine the frequency of each reported incident category, while the barangay field will be used to summarize the distribution and density of reported incidents across administrative areas.

After retrieving the report dataset, the dashboard module will group the records according to the selected visualization attribute and compute the total number of incidents for each category or time interval to generate summary datasets. For the incident timeline, reports will be arranged chronologically to illustrate the progression of emergency incidents over the selected monitoring period. The hourly, daily, and monthly frequency charts will be produced by grouping reports according to their corresponding time intervals and counting the number of incidents within each interval. Similarly, the chart showing the distribution of incidents will classify the reports based on the incident classification criteria, whereas the incident density per barangay chart will sort the reports based on the barangay wherein the incident was committed, allowing the command center to pinpoint areas that have the highest number of incidents reported.

The data sets will be transformed into JSON objects that can be utilized by the selected charting library, and shown as line graphs, bar graphs, and pie charts through the dashboard module. When there are updates due to a new report received or when filtering time, the dashboard module will automatically pull out the data set and generate statistics.

2.3.2.4. Report Records

All incident reports received by the command center will be consolidated and displayed at a reports page where it can be easily accessed through search and filtering mechanisms. Additionally, the system will allow exporting of these reports into a .csv, .xlsx or .pdf file.

2.3.3 Location Tracking Module

The location tracking module will be responsible for the beaconing of responder coordinates from the responder mobile application to the command center web application using the APRS protocol. The responders will beacon their location to the command center at five-minute intervals, the command center will then visualize these location data in a tracking interface in the web application. The responder has to manually enable or disable the beacon functionality as turning it on would block the responder from receiving radio signals which would render the messaging module to be temporarily disabled.

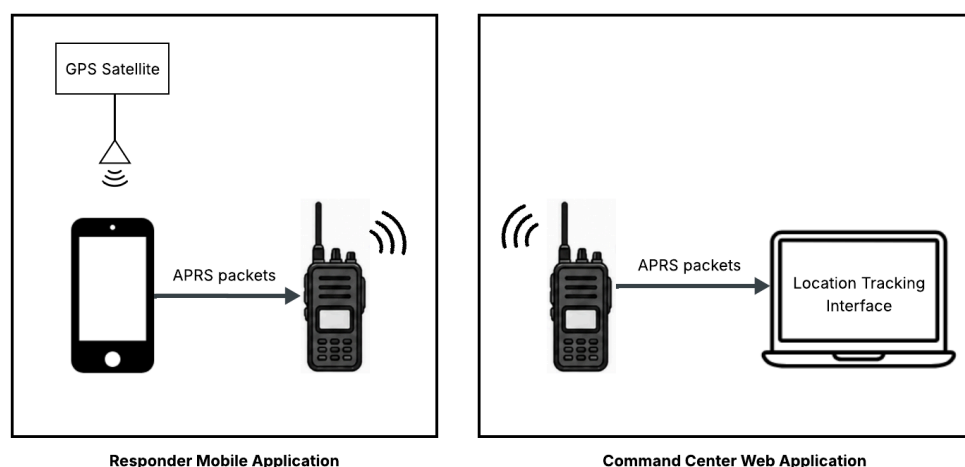


Figure 27. Location Tracking Module Flow

2.3.4 Image Enhancement Module

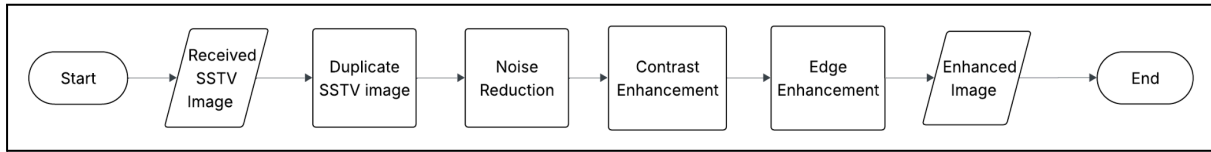


Figure 28. Image Enhancement Process Flow

The image enhancement module will be responsible for improving the visual quality and interpretability of images received at the command center after SSTV transmission. Since the SSTV works in a narrow 3 kHz bandwidth, the images that are transmitted through radio waves may be prone to forms of degradation, such as noise, low contrast, blurred edges, loss of detail, and smearing due to any signal interference or obstruction along the path of transmission. Given that one or more of these image degradations may occur individually, along with the possibility of the simultaneous occurrence of all these image degradations during SSTV transmission, a single image enhancement technique may not be sufficient to adequately improve the received image. To solve this, the image enhancement module will implement and compare multiple classical image processing pipelines by sequentially using multiple image processing techniques to handle the effects of degradations caused by SSTV transmission. Classical image processing techniques were selected because they require lower computational resources while still providing effective enhancement for SSTV images, making them suitable for offline operation and practical deployment on the command center side of the system. This approach is based on the established image enhancement principles, where image enhancement is performed progressively through noise reduction, contrast enhancement, and edge enhancement (Gonzalez & Woods, 2018; Qi et al., 2021).

As illustrated in Figure 28, the enhancement process will begin when the command center operator selects a received image for enhancement. The chosen image will be duplicated first to preserve the original image for comparison and documentation. The duplicated image will then go through a three-stage enhancement pipeline of noise reduction, contrast enhancement, and edge enhancement. Different combinations of classical image processing techniques will

be employed within these stages to come up with the best method for enhancing the received images.

The first stage of the pipeline will perform noise reduction using Median, Gaussian, and Wiener filter to help remove image distortions while preserving important image information. Median filter will be evaluated since it is highly effective in removing impulse noise from an image by replacing the pixels in the image with the median values of their neighboring pixels without affecting image edges (Qi et al., 2021). Gaussian filter will also be explored since it is found effective in suppressing high-frequency random noise through Gaussian smoothing and reducing uniformly distributed noise in an image (Qi et al., 2021). Lastly, Wiener filter will be investigated since it restores an image through minimizing the mean squared reconstruction error while considering the image degradation and noise in the image, making it suitable for images affected by blur (Gonzalez & Woods, 2018).

The second stage will perform contrast enhancement using Contrast Limited Adaptive Histogram Equalization (CLAHE). As compared to traditional histogram equalization, CLAHE improves the local contrast of an image through dividing an image into smaller regions, preventing contrast amplification in the regions from being too high (Mishra, 2021). Since SSTV-transmitted images may be prone to having uneven illumination and low contrast after the process of transmission, this technique will primarily be used within the image processing pipeline.

Finally, the third stage will perform edge enhancement to enhance the visibility of important image features that may have become blurred due to transmission or prior enhancement procedures. Unsharp masking will be evaluated since it sharpens image details by emphasizing edges while still retaining natural image appearance which makes important scene details easier to distinguish (Song et al., 2022). Laplacian sharpening will also be tested as it sharpens image edges by applying a second-order derivative operator that emphasizes fast changes in pixel intensity which makes fine structure details more visible (Song et al., 2022).

In order to determine the best enhancement approach, several enhancement pipelines based on these classical image processing techniques will be implemented and evaluated. All pipelines will consist of the same sequence of noise reduction, contrast enhancement, and edge enhancement and with just varying techniques applied within the noise reduction and sharpening stages. The outputs produced by each image pipeline will be compared to the original pre-transmission image using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). The goal of this comparison is to determine whether the enhancement process actually makes the SSTV-received images sufficiently usable for situational assessment at the command center. PSNR measures how closely the enhanced image matches the original pre-transmission image by expressing the ratio between the maximum possible pixel value and the reconstruction error on a logarithmic scale, where higher values indicate better reconstruction quality and less pixel-level distortion. Evaluation can be carried out by applying PSNR to both the raw received image and the enhanced image to see if the image enhancement module can produce a measurable improvement in the accuracy of the pixels' values when compared to the unprocessed received image.

$$PSNR = 20 \log_{10} \frac{255}{\sqrt{MSE}}$$

Formula 1. Peak Signal-to-Noise Ratio (Maharani et al., 2024)

In addition to PSNR, the Structural Similarity Index Measure (SSIM) will be used to evaluate whether the enhancement produces an image that is more perceptually similar to the original from a human visual standpoint. Unlike PSNR, which treats all pixel differences equally regardless of where they appear in the image, SSIM compares local patterns of luminance, contrast, and structure between the enhanced output and the original pre-transmission image, reflecting how a human observer would perceive the quality difference. An SSIM value closer to 1 indicates that the enhanced image is more visually faithful to the original. Using SSIM alongside PSNR ensures that the evaluation captures both numerical accuracy and perceptual quality, giving a more complete picture of whether the image

enhancement module produces results that are genuinely more interpretable for command center personnel during situational assessment.

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

Formula 2. Structural Similarity Index Measure (Wang et al., 2004)

The selected pipeline will be the one that provides the best overall balance between PSNR, SSIM, and visual interpretability. Once identified, the selected pipeline will be permanently integrated as the default enhancement method in the image enhancement module of the command center web application. The improved image will then be stored in the command center database and displayed on the monitoring dashboard alongside the original received image so that personnel can use the best available version for situational assessment.